

Lokesh Kudipudi

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EDUCATION

- Indian Institute of Information Technology Sri City** Chittor, India
Bachelor of Technology – Computer Science and Engineering CGPA: 9.13/10 2023 – 2027

SKILLS SUMMARY

- Languages:** Python, JavaScript, TypeScript, C++, HTML/CSS
- Frontend & Backend:** React.js, React Native, Tailwind CSS, Node.js, Express.js, Flask, FastAPI
- AI/ML:** PyTorch, Scikit-learn, Hugging Face Transformers, OpenCV, LangGraph, Whisper ASR, Kokoro TTS, IndicTrans2
- MLOps & Infra:** MLflow, DVC, Airflow, Docker, Grafana, Prometheus, DagsHub, Weights & Biases
- Databases & Auth:** MongoDB, Redis, JWT, Google OAuth 2.0, Socket.IO
- Tools:** Git, GitHub, Gradio, Gemini API, MCP

PROJECTS

- Mockly — Real-Time Agentic AI Interview Platform** [Github]
React, Flask, Socket.IO, Faster-Whisper, Kokoro TTS, Google Gemini, MongoDB, TypeScript
 - Cut **end-to-end time-to-first-audio by 77%** (6.6s → 1.5s; p95 6.7s → 1.4s) by pipelining streaming Gemini token output through a sentence-buffered Kokoro TTS consumer over a producer/consumer queue; instrumented a per-stage latency telemetry system (VAD / ASR / LLM / TTS) persisting p50/p95 timings to MongoDB and surfacing them in a session-level performance dashboard that isolated **LLM + TTS as ~85% of turn latency**.
 - Built a browser-side AudioWorklet capture pipeline with **RMS-based automatic barge-in** (4-frame threshold, 1.2s cooldown) enabling **mid-sentence AI interruption within one audio frame**; designed a Gemini-backed scoring service with Pydantic-typed response schema and a self-correcting interviewer that injects the last five sessions' questions as a **do-not-repeat block** into subsequent prompts.
- Dispatch — Agentic HR Outreach & MLOps System** [DagsHub]
LangGraph, FastAPI, MLflow, DVC, Apache Airflow, Grafana, Prometheus, React, TypeScript
 - Built an **end-to-end HR outreach agent** using LangGraph + Gemini that scrapes LinkedIn #hiring posts via a persistent Playwright profile, normalizes contact data, and generates personalized email drafts with a **hard human-approval gate** before pushing to Gmail; secured with async FastAPI + SQLAlchemy/Alembic backend and a React/TypeScript frontend with typed API contracts.
 - Delivered a **full MLOps backbone** with MLflow on DagsHub for experiment tracking and metric logging, DVC for **dataset versioning enabling reproducible evaluations by commit**, and Airflow-orchestrated pipelines monitored via Grafana and Prometheus dashboards.
- Railway Sleeper Crack Detection — Computer Vision**
YOLOv8, OpenCV, Weights & Biases, Python
 - Engineered a **two-stage inference pipeline** — YOLOv8n-seg for sleeper-region segmentation (mAP50: 94.9%, mask precision: 97.2%), followed by tiling each ROI into 1024×1024 patches for crack detection — lifting **crack mAP50 by +325%, precision by +272%, and mAP50-95 by +363%** over the direct-downscaling baseline; benchmarked YOLOv8n, v11n, and v12n head-to-head.
 - Manually annotated **421 crack instances and 100+ segmentation masks** across 444 high-resolution real-world images; reduced **train-validation overfitting gap from ~45% to ~17%** via mosaic, copy-paste augmentation, and increased weight decay, establishing a reproducible regularization recipe for small-dataset detection tasks.

ACHIEVEMENTS

- GATE 2026 – Computer Science:** All India Rank **1308** with a score of **689**

POSITION OF RESPONSIBILITY

- Epoch AI-ML Club** IIIT Sri City
Club Lead (promoted from Deep Learning Domain Core) August 2024 – April 2026
 - Promoted from domain core member to Club Lead within a year; spearheaded technical workshops on Deep Learning and Generative AI, reaching 100+ students.
 - Built and deployed event and club websites, and co-organized inter-college AI/ML competitions and speaker sessions.

CERTIFICATIONS

- The Complete JavaScript Course 2023 (Udemy):** From Zero to Expert
- The Ultimate React Course 2023 (Udemy):** React, Next.js, Redux & More
- Deep Understanding of Deep Learning (Udemy):** ANN, CNN, RNN, GANs, Autoencoders, Transfer Learning
- Machine Learning A-Z (Udemy):** Regression, Classification, Clustering, NLP, Deep Learning